

NATIONAL CONNECTIVITY Via LoRaWAN In Just Eight Months



Saurabh Durgapal
is working as
technology
journalist at EFY

Google trends graph shows a rapid increase in the search for Long Range Wide Area Network (LoRaWAN) since April 2015. "Initial challenges have been in making the common person understand the use cases. Post that, public response has been overwhelming," said Rishabh Chauhan, global community manager, The Things Network, in an earlier interaction with EFY. It seems to have turned into a global phenomenon ranging over some very interesting applications.

We were able to find one use case in the Indian subcontinent as well. Using LoRaWAN, engineers solved the problem of indication for chlorine gas leakage in power plants. Chauhan says, "A lot of people are using the technology and trying to evaluate which use case works for them, and eventually adopting the network." Turns out, it makes a lot of sense to implement LoRa in real-life problems.

However, Krupa Shukla, CEO, Yups Tech solutions, highlights some issues, "LoRa is becoming popular because of its low cost and low power consumption, but its communication architecture is still at an evolving stage."

Developments in LoRaWAN

Using frequencies lower than 2.4GHz or 5.8GHz enables much better coverage, especially when the nodes are inside buildings. Sub-1GHz frequencies are normally used; however, the technology is frequency agnostic.

LoRa wireless system makes use of unlicensed frequencies available worldwide.

Commonly-used bands include 868MHz for Europe, 915MHz for North America and 433MHz band for the Asian subcontinent. However, another classification could be spread spectrum versus narrow-band system.

Spread spectrum through the years

The secrets of spread spectrum were not unknown to the US military. Since spread spectrum is below noise level, it is neither decipherable nor it can be jammed, intentionally or unintentionally. Even though the jamming noise can be introduced, lack of a key-set does not allow it to spread and mix with the original signal, thereby jamming it without any effect. It is immune to major atmospheric conditions as well.

Since World War II the US military have been deploying spread spectrum for their torpedoes and missiles to hide them from deciphering, only at great cost. The cost has reduced significantly and, in the recent years, companies have released hardware compatible with spread spectrum signals at a very low cost.

What spread spectrum lacks. In comparison, Narrowband IoT (NB-IoT) has advantages of an already mature ecosystem for mobile networks with support from telecom equipment vendors. "Long Term Evolution also has an advantage here, as no time and cost has to be invested in the infrastructure, comparatively," adds Shukla.

NB-IoT technology standardised by 3GPP standards body is a narrow-band radio technology specially designed for the IoT. With capabilities like indoor coverage, low cost, long battery life and support for a large number of devices, this technology can be deployed in Global System for Mobile communication and Long Term Evolution spectrum. It uses licensed spectrum to provide secure and stable services.

Fig. 1: LoRa receiver used in the chlorination plant project

